

RETAIL DEMAND INTELLIGENCE: FORECASTING & DEMAND DRIVER ANALYSIS

Inventory Planning, Pricing Dynamics
& Operational Performance



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly &
ML libraries

PRESENTED BY

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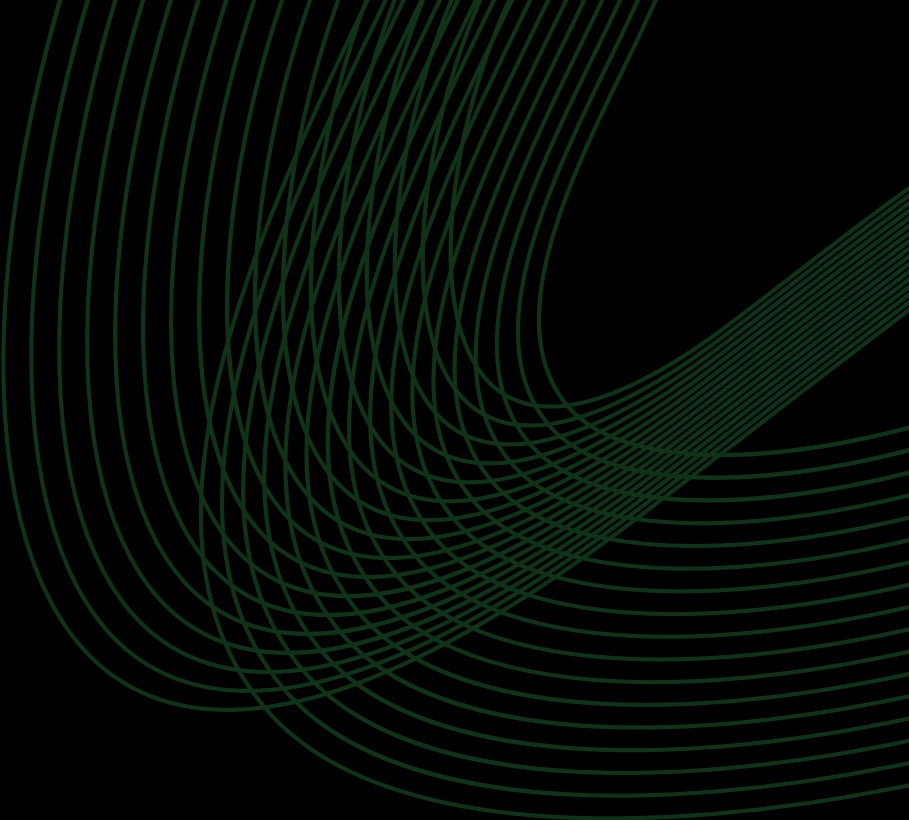


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1. OVERVIEW & BUSINESS OBJECTIVE

1.1 Project Overview

Managing Google Cloud Platform resources across multiple services, regions, and workloads is complex, with diverse usage patterns and cost drivers. This makes it challenging to identify the primary factors influencing cloud expenditure, resource efficiency, and opportunities for targeted cost optimization.

Objective:

To analyze the Google Cloud Platform usage and cost dataset to uncover patterns, insights, and trends that can help businesses make data-driven decisions around resource efficiency, cost optimization, and cloud expenditure management.



1. OVERVIEW & BUSINESS OBJECTIVE

1.1 Project Overview

Retail businesses operate in a dynamic environment where demand is influenced by factors such as historical sales, inventory, pricing, promotions, seasonality, and region.

Accurate demand estimation helps businesses maintain optimal inventory, reduce stockouts and excess stock, and improve procurement and operational planning.

The Retail Demand Intelligence project uses historical retail data and machine learning to predict demand and identify the key factors influencing it.

The project combines:

- Data understanding and cleaning
- Exploratory Data Analysis
- Feature engineering
- Machine learning regression
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented insights

The objective is not only to predict demand accurately, but also to understand why the model makes its predictions & how these insights can support better decision-making.



1.2 PROBLEM STATEMENT

Retail demand is influenced by multiple interacting variables, making manual forecasting or reliance on historical averages insufficient for many situations.

The key problem addressed in this project is:

How can historical retail data be used to build a machine learning model that accurately predicts product demand while also identifying the major factors driving demand?

The project therefore treats Demand as the target variable and uses historical, operational, pricing, promotional, environmental, and temporal information as predictive features.



1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Develop a machine learning-based retail demand prediction system.
- Understand the underlying structure and characteristics of the dataset.
- Identify important patterns through exploratory data analysis.
- Create meaningful temporal, inventory, pricing, and historical features.
- Train and compare multiple regression models.
- Evaluate model performance using appropriate regression metrics.
- Investigate the largest prediction errors.
- Identify the most influential demand-driving features.
- Identify limitations and possible improvements for future development.

